Umrah Guide Knowledge Base

Computer Science and Engineering
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1. Problem Title

Development of an Umrah Guide Knowledge Base using Prolog.

2. Problem Statement

Umrah is one of the most significant acts of worship in Islam, involving a series of rituals performed in Makkah. Many pilgrims, especially first-timers, struggle to remember the correct sequence of rituals, the rules of Ihram, the duas for each occasion and guidance specific to their condition or gender.

This project presents a Prolog-based Knowledge Base that serves as an intelligent Umrah guide. The system stores and reasons over structured Islamic knowledge, allowing users to query information interactively.

Key Components:

1. **Places**
   - Stores 8 significant Umrah locations including Masjid al-Haram, Kaaba, Hateem, Zamzam Well, Safa, Marwa, Meeqat, and Masjid Aisha with descriptions and categories.
2. **Rituals in Sequence**
   - All 8 steps of Umrah are stored in order. The system answers what comes before or after any ritual.
3. **Ihram Rules Engine**
   - Rules about what is allowed and forbidden during Ihram. Users receive immediate ALLOWED or FORBIDDEN responses with reasons.
4. **Dua Knowledge Base**
   - Specific duas for 7 key occasions during Umrah with Arabic text and English meaning.
5. **Pilgrim Advisor**
   - Personalized guidance based on the pilgrim's gender and physical condition. Special concessions for elderly, pregnant, and disabled pilgrims.

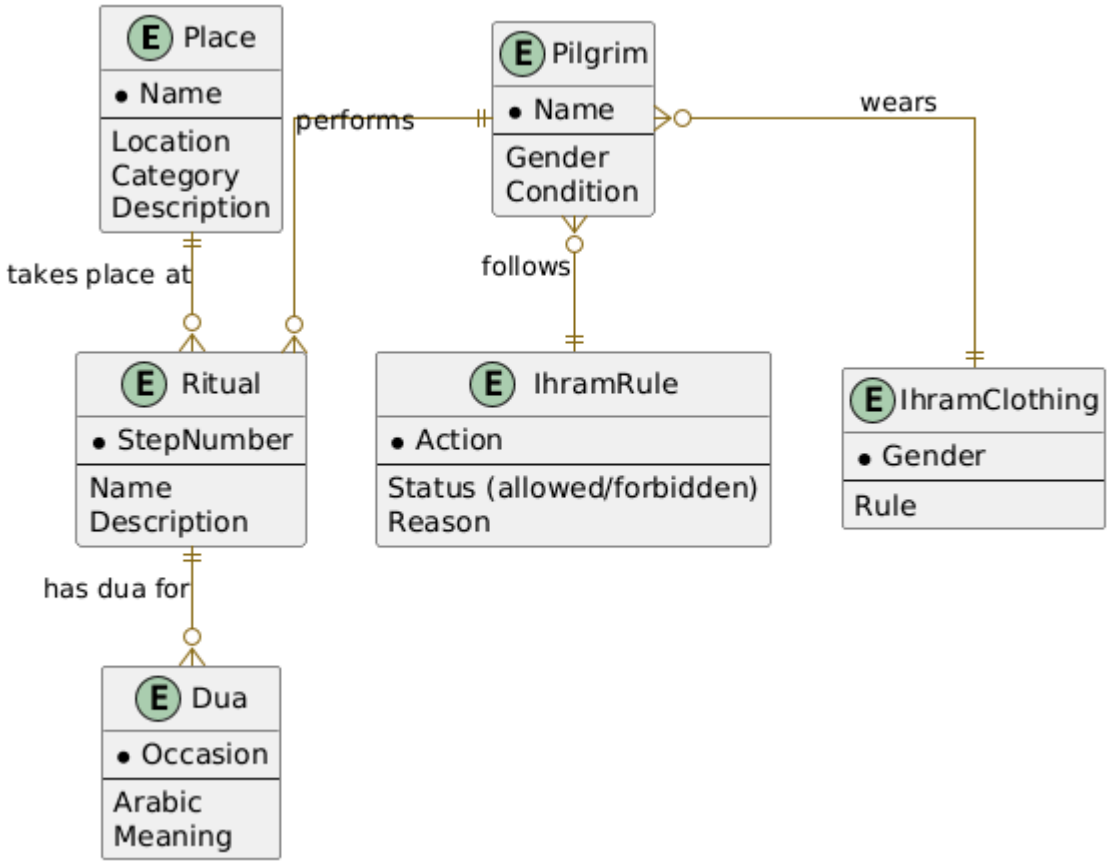
3. Tools and Languages Used

- **Programming Language:** SWI-Prolog (version 10.0.2) - used for knowledge representation, fact storage, rule definition and interactive querying.
- **Development Environment:** SWI-Prolog IDE - interactive console for loading, testing and debugging the knowledge base.
- **Text Editor:** Notepad (Windows) - used to write and edit the Prolog source code file.
- **Operating System:** Windows 11.

- **Version Control:** GitHub - repository available at https://github.com/lamiatabassumorpa/AI-lab

4. ERD Diagram

The Entity-Relationship Diagram below represents the key entities and their relationships in the Umrah Guide Knowledge Base.



Entities:

1. Place — Name (PK), Location, Category, Description
2. Ritual — StepNumber (PK), Name, Description
3. IhramRule — Action (PK), Status (allowed/forbidden), Reason
4. Dua — Occasion (PK), Arabic, Meaning
5. Pilgrim — Name (PK), Gender, Condition
6. IhramClothing — Gender (PK), Rule

Relationships:

- Pilgrim performs Ritual
- Pilgrim follows IhramRule
- Pilgrim wears IhramClothing
- Ritual takes place at Place
- Ritual has Dua

5. Knowledge Base Design

The knowledge base is built using two core building blocks of Prolog — Facts and Rules.

- A Fact is a piece of information that is unconditionally true. Example: place(kaaba, inside_masjid_al_haram, structure, 'The cubic structure, Qibla of all Muslims').
- A Rule is a logical condition that derives new information from existing facts. Example: inside_haram(X) :- place(X, inside_masjid_al_haram, _, _).

Section 1 — Places

Stores 8 significant Umrah locations using the predicate place(Name, Location, Category, Description). Rules allow querying all holy places, places inside the Haram, descriptions and categories.

Section 2 — Rituals in Sequence

Stores the 8 steps of Umrah in order using ritual(StepNumber, Name, Description). Rules allow finding next/previous rituals, retrieving descriptions and identifying the first and last ritual.

Section 3 — Ihram Rules Engine

Stores 8 allowed and 9 forbidden actions using allowed_in_ihram(Action, Reason) and forbidden_in_ihram(Action, Reason). The check_action rule returns a clear ALLOWED or FORBIDDEN response with the reason.

Section 4 — Dua Knowledge Base

Stores 7 duas using dua(Occasion, Arabic, Meaning). The show_dua rule displays the occasion, Arabic text and English meaning in a readable format.

Section 5 — Pilgrim Advisor

Stores 5 pilgrims with different genders and conditions using pilgrim(Name, Gender, Condition). Rules provide personalized guidance based on each pilgrim's profile. Gender-specific Ihram clothing rules are also included.

6. Sample Input / Output

The following queries were tested in SWI-Prolog after loading the knowledge base.

Query 1 — List all holy places:

?- holy_place(X).

```
SWI-Prolog console
File  Settings  Tools  Debug  Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

23 ?- consult(['c:/Users/User/OneDrive/Desktop/22101184.pl']).
true.

24 ?- holy_place(X).
X = masjid_al_haram ;
X = kaaba ;
X = hateem ;
X = zamzam_well ;
X = safa ;
X = marwa ;
X = meeqat ;
X = masjid_aisha.
```

Query 2 — Find next ritual after Tawaf:

?- next_ritual(tawaf, X).

```
32 ?- next_ritual(tawaf, X).
X = two_rakah_prayer.
```

Query 3 — Check actions in Ihram:

?- check_action(cut_nails).

```
40 ?- check_action(cut_nails).
FORBIDDEN: Trimming nails is forbidden in state of Ihram
true .
```

?- check_action(recite_quran).

```
41 ?- check_action(recite_quran).
ALLOWED: Recitation of Quran is encouraged in Ihram
true .
```

Query 4 — Show dua:

?- show_dua(drinking_zamzam).

```
49 ?- show_dua(drinking_zamzam).
Occasion : drinking_zamzam
Arabic   : Allahumma inni asaluka ilman nafia
Meaning  : O Allah I ask You for beneficial knowledge and wide sustenance
true .
```

Query 5 — Show guidance for a pregnant pilgrim:

?- show_guidance(maryam).

```
56 ?- show_guidance(maryam).
Pilgrim   : maryam
Gender    : female
Condition : pregnant
Guidance  : Consult a doctor before performing Umrah. Tawaf can be done at outer areas. Rest frequently.
true .
```

Query 6 — Show Ihram clothing rules:

?- show_ihram_clothing(male).

```
59 ?- show_ihram_clothing(male).
Gender   : male
Clothing: Two unstitched white cloths, no head covering, no stitched clothing
true.
```

Tried to keep at least one query from each section.

7. Conclusion

This project successfully implements an Umrah Guide Knowledge Base using SWI-Prolog. The system effectively organizes and reasons over Islamic knowledge related to the performance of Umrah, making it accessible and queryable in an interactive manner.

The knowledge base covers five key areas — holy places, ritual sequences, Ihram rules, duas and pilgrim guidance -- each built using Prolog facts and rules. Users can query the system to find the correct sequence of rituals, check

what is allowed or forbidden in Ihram, retrieve duas for specific occasions and receive personalized guidance based on their gender and physical condition.

The use of Prolog's logical inference capabilities makes this knowledge base a powerful tool for rule-based reasoning. Unlike a traditional database, this system does not just store data -- it reasons over it and provides intelligent responses to queries.

This project also holds personal significance, as it was inspired by a recent Umrah visit, making the knowledge represented here both authentic and meaningful.

8. Challenges

1. **Learning Prolog from Scratch**
This was the first time working with Prolog, which required understanding a completely different programming paradigm. Unlike conventional languages like Python or Java, Prolog is based on logic and facts rather than instructions and loops.

2. **Understanding Backtracking and the Semicolon**
Learning that pressing ; triggers backtracking to find the next solution was a key concept that had to be understood before queries could be used effectively.

3. **Writing Rules Correctly**
Writing rules using variables like X, D, N, and N1 was initially challenging. Understanding that capital letters represent variables and that _ is a wildcard required careful practice.

4. **Handling Unknown Actions in check_action**
Ensuring that check_action handled actions not found in the knowledge base required the use of the \+ operator (negation). Understanding how to use negation in Prolog was resolved through testing.

5. **Structuring the Knowledge Base**
Organizing five different sections in a single .pl file while keeping facts and rules clean, readable, and logically consistent required careful planning and multiple revisions.